



Let the mind manage the body
Que l'esprit gère le corps

SCIENCE

(Subject code No. P141/1)

Index Number:

MAURITIUS EXAMINATIONS SYNDICATE
Primary School Achievement Certificate Assessment
Grade 5 Modular
October 2018
Time 1 hour Total Marks: 50

INSTRUCTIONS TO CANDIDATES

1. Check that this assessment booklet contains 6 questions printed on 10 pages numbered 2, 3, 4, 5, 6, 7, 8, 9, 10 and 11.
2. Write your Index Number on the assessment booklet in the space provided above.
3. You should not use red, green or black ink in answering questions.
4. Write all your answers clearly in the assessment booklet.
5. Attempt all questions.

Question	Marking		Revision		Control	
	Marks	Sig	Marks	Sig	Marks	Sig
1						
2						
3						
4						
5						
6						
Total						
Sig (HoG)						

Question 1 (5 marks)

Circle the letter which shows the correct answer.

1. **Diagram 1** shows a man skiing down the icy slope of a mountain.

Two **properties** of ice that allow the man to slide easily are that ice is

- A** warm and soft
- B** warm and slippery
- C** hard and slippery
- D** cold and rough

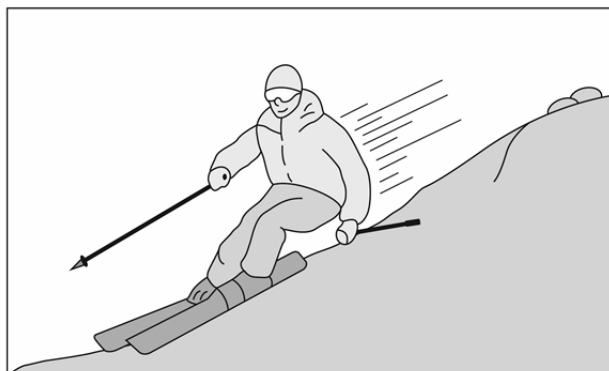


Diagram 1: A man skiing

2. **Diagram 2** shows part of a hibiscus plant.

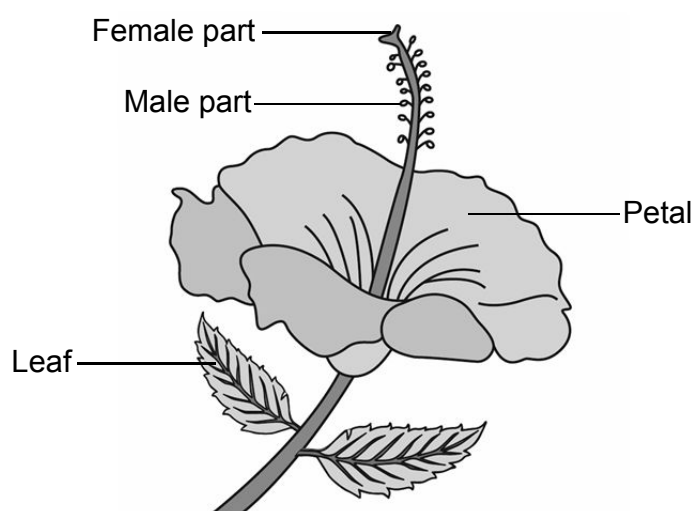


Diagram 2: Part of a hibiscus plant

Which one of the following must receive pollen for **pollination** to occur?

- A** Female part
- B** Male part
- C** Leaf
- D** Petal

3. Which **one** of the following plants has its habitat in **wetlands**?
- A Bulrush
 - B Cactus
 - C Fern
 - D Papaya
4. Which **one** of the following plants is cultivated for its **stem mainly**?
- A Tea
 - B Sugar cane
 - C Beetroot
 - D Water lily
5. **Diagram 3** shows a washing machine.

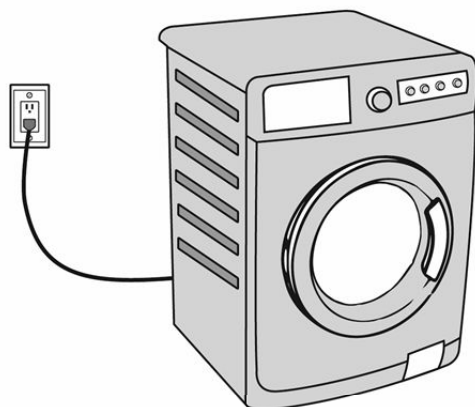


Diagram 3: A washing machine

What **main** energy transformation takes place in the washing machine when it is in use?

- A Chemical energy is transformed into electrical energy
- B Electrical energy is transformed into light energy
- C Chemical energy is transformed into sound energy
- D Electrical energy is transformed into movement energy

Please turn over this page

Question 2 (10 marks)

Marks

Diagram 4 shows a piece of electrical wire with one end connected to the negative terminal of a dry cell.

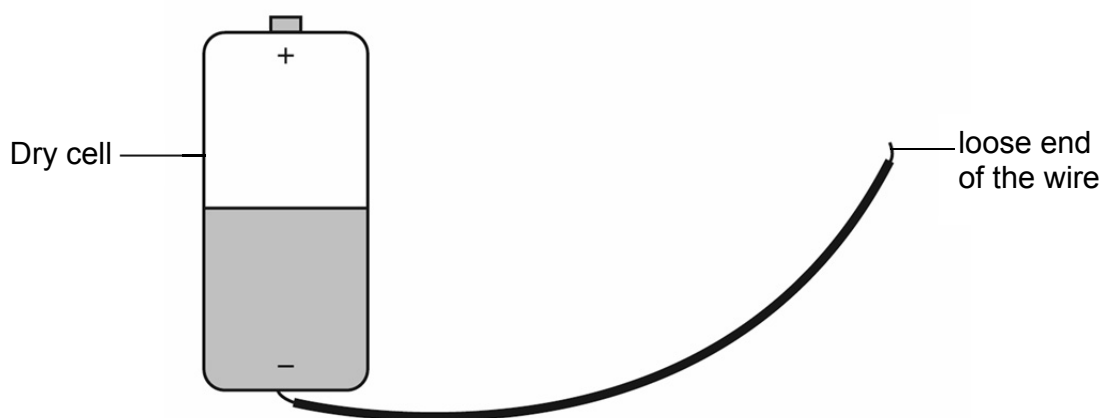


Diagram 4: Electrical wire with one end connected to a dry cell

- (a) (i) Does **Diagram 4** show a **closed** or an **open** circuit?

[1]

- (ii) Explain your answer to **part (a)(i)**.

[1]

- (b) (i) Will the circuit be **closed** or **open** if the loose end of the wire is connected to the positive terminal of the dry cell and the other end is connected to the negative terminal?

[1]

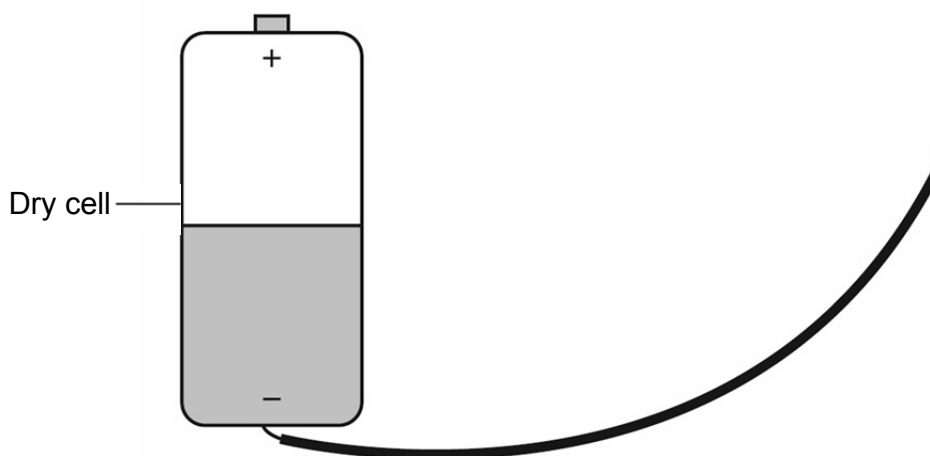
- (ii) Explain your answer to **part (b)(i)**.

[1]

- (c) (i) Name a **component** that can be added to the circuit to show that electricity is flowing.

[1]

- (ii) Complete the circuit below by **drawing** the **component** you gave in **part (c)(i)** to show that electricity is flowing in the circuit.



[2]

- (d) What **main** energy transformation takes place in the completed circuit in **part (c)(ii)**?

_____ energy in the **dry cell** is transformed into

_____ energy in the **added component**.

[2]

- (e) Name a **component** that can be used to open or close the circuit to save energy in the dry cell.

[1]

Question 3 (11 marks)

Marks

- (a) Match each part of a plant to its function.

Part of a plant
Leaf
Stem
Root
Fruit

Function
It protects and nourishes the seeds.
It takes up water and mineral salts from the soil.
It carries water from the roots to other parts of the plant.
It attracts insects.
It makes food for the plant.

[4]

- (b) Fill the blanks in the sentences below.

(i) Soil erosion happens through the action of heavy _____ and strong winds.

(ii) During soil erosion, fertile soil is carried away causing the land to become _____.

(iii) The soil that is carried away is often deposited at the mouth of _____ and affects the organisms living there.

[3]

- (c) Mangroves are trees which are planted to protect the coastal regions. How do mangroves protect the coast?

[2]

- (d) Give **one** way how an oil spill can destroy a water habitat.

[2]

Question 4 (8 marks)

Marks

- (a) Study **Diagram 5**, which shows the water cycle and a hydroelectric power station at the bottom of a waterfall.

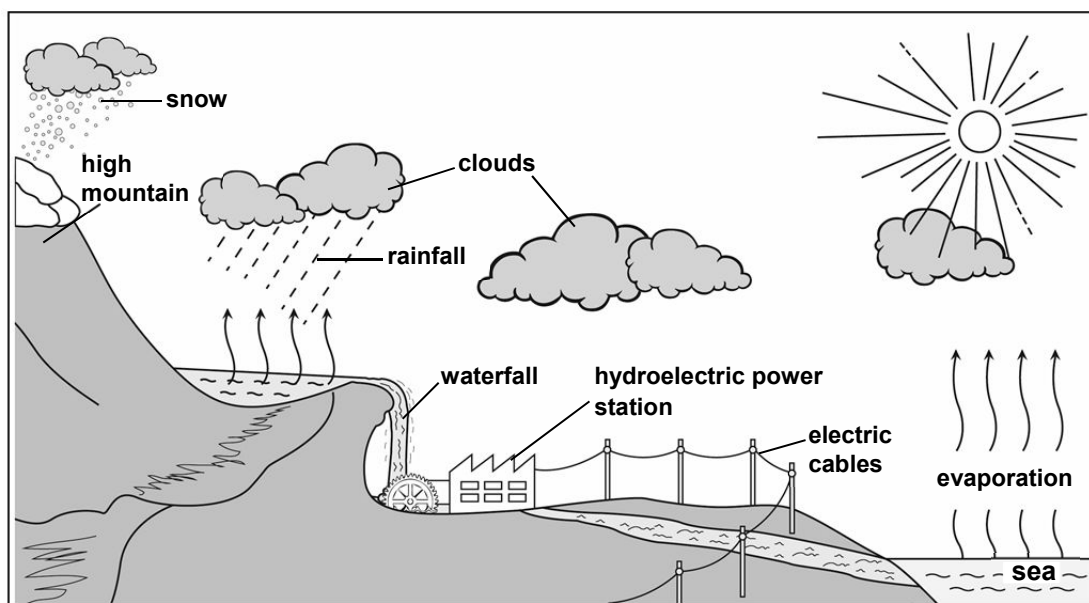


Diagram 5: Water cycle and a hydroelectric power station

- (i) What is the **source** of energy responsible for the water cycle?

[1]

- (ii) What **form** of energy is responsible for the evaporation of water?

[1]

- (iii) Fill in the blanks in the sentence below:

High up in the atmosphere, the temperature is _____
and water droplets in the clouds freeze into _____ and fall as
snow on mountain tops.

[2]

- (b) The hydroelectric power station uses the force of falling water to produce electricity.

- (i) What **energy transformation** takes place at the power station?

The _____ energy of the falling water is converted into
_____ energy.

[2]

- (ii) Electric cables are used to distribute the electricity from the power station to towns and cities.

Why are the electric cables made of copper?

[1]

- (c) In Mauritius, which **source** of energy is **mainly** used to run vehicles?

[1]

Question 5 (9 marks)

Diagram 6 shows the saddle-backed tortoise which once lived in Rodrigues. This tortoise has completely disappeared now.



Diagram 6: A saddle-backed tortoise that once lived in Rodrigues

- (a) How do we call an animal which has **completely** disappeared?

[1]

- (b) The saddle-backed tortoise could feed on leaves high up on plants. Suggest **one** way how this tortoise was adapted to live in its habitat.

[2]

- (c) Another tortoise, the 'Aldabra Giant Tortoise' from Seychelles islands, was introduced in Rodrigues.

How do we call an animal which has been introduced from another country?

[1]

- (d) **Diagram 7** shows the Pink Pigeon. The Pink Pigeon is a bird which is found only in Mauritius. It lives in the forests of Mauritius.



Diagram 7: Pink Pigeon

- (i) Give **one** way how the forests of Mauritius are a suitable habitat for the Pink Pigeon.

[2]

- (ii) The Pink Pigeon almost disappeared some years back.
Give **one** way how it can be protected.

[1]

- (iii) Why is it important to protect animals such as the Pink Pigeon?

[2]

Question 6 (7 marks)

Marks

Diagram 8 shows coconut trees in Mauritius.



Diagram 8: Coconut trees

Three children are discussing how the first coconut tree grew on the island of Mauritius.

This is what they say:

Asif: “The coconut tree grew from a leaf of the tree which floated on the seawater to the shores of the island.”

Asha: “The coconut tree grew from a coconut seed which floated to the island.”

Reshma: “The coconut tree grew from the soil without needing a leaf or a seed.”

(a) (i) Which **one** of Asif, Asha or Reshma may be correct? _____ [1]

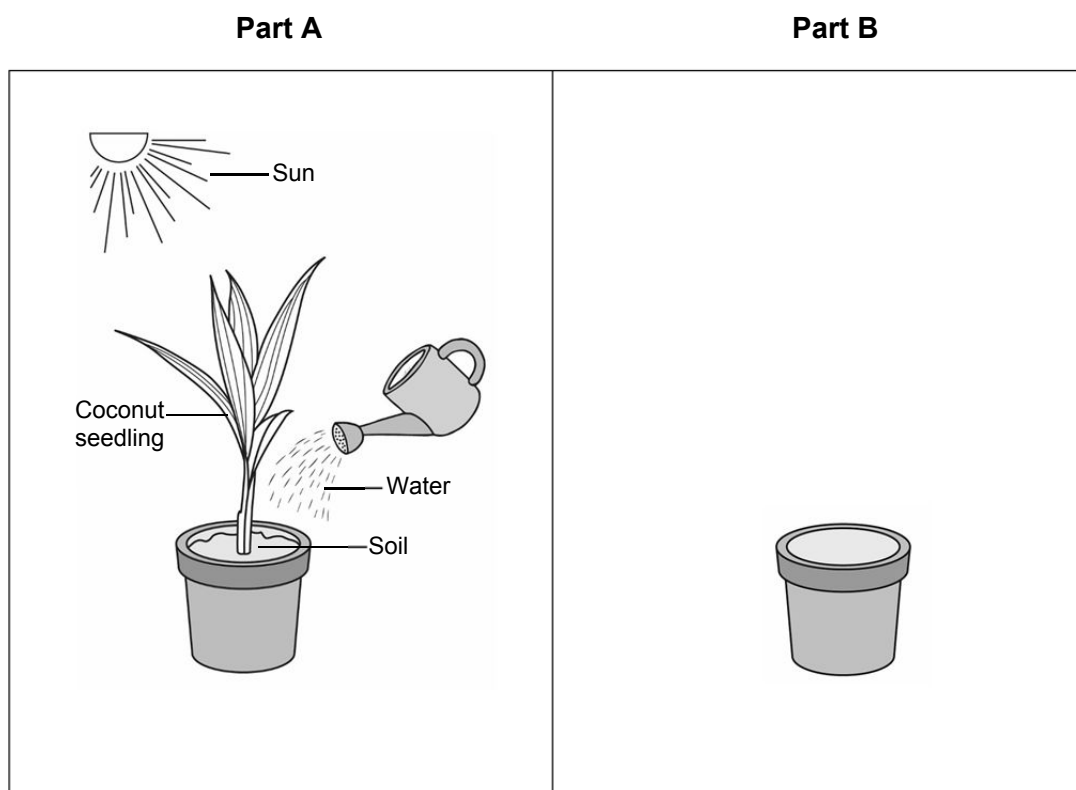
(ii) Explain your answer to **part (a)(i)**

[1]

- (b) The children carry out an experiment to find out about the **importance of water** for coconut seedlings to grow.

In **Part A**, the first part of the experiment has been drawn and labelled for you.

Draw and **label** the **second part** of the experiment that the children must carry out in **Part B**.



[2]

- (c) To compare the **coconut seedlings** in **Part A** and **Part B**, which observation can the children make?

[1]

- (d) What do you expect to happen to the coconut seedlings after some time?

Seedling in **Part A**: _____

Seedling in **Part B**: _____

[2]

BLANK PAGE